DermFusion: An Uncertainty-Aware Cross-Attention CNN–Transformer Framework for Multi-Task Skin Lesion Classification and Segmentation

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Abstract*—Melanoma is the deadliest form of skin cancer, yet early-stage disease is nearly always curable, making accurate automated detection a high-value clinical target. State-of-the-art deep-learning systems for dermoscopy analysis typically treat classification and segmentation as separate tasks and produce insufficient confidence scores, potentially limiting their safe deployment. We proposed *DermFusion*, which integrates both within a single ResNet-50+CBAM/ViT-B/16 dual-backbone network connected by a bidirectional cross-attention fusion layer. A U-Net decoder with Focal-Tversky deep supervision handles boundary prediction, while a Monte Carlo dropout head quantifies predictive uncertainty. Post-hoc temperature scaling calibrates model confidence, ensuring predicted probabilities better match actual accuracy. On the HAM10000 dataset, DermFusion achieves 87.82% accuracy, 0.9799 ROC-AUC, 0.9359 Dice, and 91.2% melanoma sensitivity. Calibration improves significantly, with the expected calibration error decreasing from 0.4395 to 0.1575, representing a 64.1% relative reduction. It has been observed that when an uncertainty-based referral threshold applies, 68% of cases classified autonomously reach 96.4% accuracy, with the remaining 32% flagged for expert review. external robustness evaluation on PH², ISIC 2019, and PAD-UFES-20 datasets was applied, which shows that the combined representation works well across different acquisition domains. Ablation, fairness, and failure-mode studies characterize where and why the proposed *DermFusion* system succeeds or breaks down.

***Index Terms*—Skin cancer, melanoma, vision transformer, cross-attention CNN–transformer, monte carlo dropout.**

I. Introduction

MELANOMA accounts for the majority of skin cancer deaths despite making up less than 2% of diagnoses, yet five-year survival for stage-I disease exceeds 98%. The gap between early and late detection is therefore among the steepest in oncology. Catching melanoma early is almost always curative, but missing or delayed detection typically is not. This stark asymmetry has driven sustained interest in computer-aided dermoscopy analysis as a tool to extend specialist-level diagnosis to contexts where dermatologist access is limited. Dermoscopy illuminates subsurface lesion features, pigment networks, vessels, border structure, and regression that the naked eye cannot see. Its diagnostic value is well established, yet even experienced dermatologists disagree on a substantial fraction of equivocal cases [1]. Automated analysis is therefore not simply a convenience but a practical necessity wherever dermatology expertise is thinly distributed.

The Convolutional Neural Network (CNN) [2]–[4] brought strong classification results to dermoscopy, but its limited ability to model global contextual relationships, loss of spatial information, and long-range dependencies make it challenging to model lesion-level global structure. Vision Transformers [5] address the global modeling gap through self-attention, at the cost of the translation-equivariant local texture cues that CNNs handle well. Beyond architectural advances, existing dermatology AI systems inadequately address three critical deployment-related challenges essential for safe clinical implementation. These are(i) joint classification-segmentation optimization, (ii) calibrated confidence scores that mean what they say probabilistically, and (iii) principled uncertainty triage so the system can say, ‘"I am not sure," and defer to a clinician rather than guess [6], [7]. To address all these three challenges simultaneously, we have proposed *DermFusion*. Unlike previous hybrid dermatology models focused primarily on feature fusion, DermFusion integrates calibration-aware uncertainty estimation, segmentation-guided learning, deep supervision, selective prediction triage, and fairness-aware evaluation into a unified framework designed to enhance clinical reliability, robustness, and deployment readiness. The complete conceptual architectural overview is presented in Figure 1, and all mathematical symbols used throughout the paper are defined in Table II. The primary contributions of this work are:

- DermFusion, a novel dual-backbone CNN–Transformer framework that integrates ResNet-50+CBAM and ViT-B/16 via bidirectional cross-attention fusion for simultaneous skin lesion classification and segmentation, improving feature representation, prediction reliability, and confidence-aware clinical deployment.
- An uncertainty- and calibration-aware decision framework

incorporating Bayesian MC Dropout, variance-based uncertainty estimation, temperature scaling, and threshold-sensitive triage analysis, achieving a 64.1% reduction in expected calibration error while enabling reliable confidence-based referral decisions through quantified coverage–accuracy trade-offs.
- Establishes the robustness and clinical reliability of DermFusion through extensive 5-fold cross-validation;external robustness evaluation on PH², ISIC 2019, and PAD-UFES-20 datasets; and comprehensive ablation, fairness, and failure-mode analyses to demonstrate strong cross-domain generalization and transparency characterization of model behavior under diverse real-world conditions.

II. Related Work

Recent advances in automated skin lesion analysis have evolved through three closely interconnected research directions: deep learning-based skin lesion classification, transformer and hybrid medical image analysis, and uncertainty-aware trustworthy artificial intelligence. CNNs were used to extract local texture and morphological features for lesion classification, but their limited ability to capture long-range contextual relationships motivated the adoption of ViTs and hybrid CNN–Transformer architectures, which combine local feature extraction with global contextual understanding. Now research has shifted beyond predictive accuracy toward clinically reliable deployment by incorporating uncertainty quantification, confidence calibration, and selective prediction mechanisms. These advancements enable models not only to classify lesions accurately but also to estimate prediction confidence and identify uncertain cases requiring expert review.

Esteva et al. [8] showed CNNs can match dermatologists on a binary benign/malignant task, a landmark result that opened the field. The HAM10000 benchmark [9] then shifted focus to seven-class multi-lesion discrimination; ResNet [2], DenseNet [3], and EfficientNet [4] all became standard baselines. Channel-spatial attention modules such as CBAM [10] were subsequently introduced to refine discriminative CNN feature responses by suppressing background channels and spatial regions; DermFusion adopts CBAM. The ConvNeXt family [11], [12] extended these advances with modernized purely convolutional architectures. The ISIC challenge series [13], [14] provides standardized community benchmarks that are used in this work for external validation.

Transformer architectures were adapted for computer vision through Vision Transformers [5], whose self-attention formulation became the ViT backbone. Although ViTs effectively capture long-range global dependencies, their limited translation invariance compared to CNNs has motivated the development of hybrid architectures. A significant advancement in hybrid medical imaging was introduced by Chen et al. [15] through TransUNet, which effectively combined CNN-based feature extraction with Transformer-based global context modeling for segmentation tasks. Cao et al., in their paper [16], introduced Swin-Unet, which leverages shifted-window self-attention for hierarchical feature learning. Xie et al. [17] proposed SegFormer, which is an efficient Transformer-based framework designed for semantic segmentation. IWu et al. [18] developed FAT-Net with feature-adaptive transformers for skin lesion segmentation, but this architecture primarily focuses on segmentation accuracy and does not address prediction uncertainty, confidence calibration, or model reliability, which are essential for clinical deployment. MedMamba [19] introduces selective scanning for efficient dependencies through selective scanning with lower computational complexity than Transformers; however, it offers limited interpretability and lacks uncertainty estimation, calibration, and clinical reliability assessments. In the paper [20], Zhou et al. proposed UNet++, which employs nested dense skip connections to reduce the semantic gap between encoder and decoder. But it increases computational complexity and parameter count, while lacking mechanisms for global context modeling, uncertainty quantification, and deployment-oriented reliability assessment.

A model that assigns high confidence to a wrong prediction is more dangerous than one that abstains. Gal and Ghahramani [21] demonstrated that maintaining dropout during inference enables approximate Bayesian inference through Monte Carlo sampling. Consequently, the variance computed across T_{mc} stochastic forward passes serves as a principled measure of predictive uncertainty and can be used to identify cases requiring clinical review. In the paper [6], Leibig et al. showed that the variance stabilizes after eight to fifteen passes, which sets our T_{mc}=10. For multi-task training, Kendall et al. [22] observed that task-specific loss magnitude carries implicit information about noise level and derived a principled weighting scheme. Guo et al. [7] demonstrated that temperature scaling, which adjusts logits using a validation-optimized scalar parameter, effectively calibrates predicted probabilities while introducing negligible computational overhead during inference. Carse et al. [23] and Gawlikowski et al. [24] comprehensively reviewed uncertainty quantification and calibration methods, highlighting their critical role in developing reliable and trustworthy AI systems. In the paper [25] Pakzad et al. demonstrated that variations in image acquisition devices can create color distribution shifts, leading to demographic bias and emphasizing the necessity of fairness-aware evaluation in medical AI. A summary of the above literature work is shown in Table I. It exhibits ResNet, EfficientNet, and ConvNeXtV2 approaches that address only diagnostic predictions but lack lesion boundary delineation capabilities. Segmentation-focused models such as FAT-Net and Swin-Unet accurately delineate lesion boundaries but lack disease classification capabilities, uncertainty quantification mechanisms, and confidence-aware prediction modeling. Although TransUNet enables joint classification and segmentation, it does not incorporate prediction calibration, uncertainty quantification, or confidence-based triage mechanisms required for trustworthy clinical deployment.

The preceding review reveals that most of the approaches address classification, segmentation, uncertainty estimation, and clinical decision support largely in isolation. To the best of our knowledge, there is still a critical research gap: existing state-of-the-art frameworks do not provide a hybrid architecture that jointly optimizes multi-class lesion diagnosis, precise lesion boundary segmentation, uncertainty-

TABLE I: Capability comparison across dermatology AI frameworks. Cls = multi-class disease classification; Seg = boundary segmentation; Calib = explicit ECE calibration; Uncert = uncertainty estimation; Triage = uncertainty-guided selective prediction.○ = partial/encoder-level only.

Method	Cls	Seg	Calib	Uncert	Triage
ResNet50 [2]	✓	–	–	–	–
EfficientNet-B4 [4]	✓	–	–	–	–
FAT-Net [18]	–	✓	–	–	–
TransUNet [15]	○	✓	–	–	–
Swin-Unet [16]	–	✓	–	–	–
MedMamba [19]	✓	–	–	–	–
DermFusion (Ours)	✓	✓	✓	✓	✓

calibrated prediction, and confidence-aware triage decision-making within trustworthy clinical deployment. To address these problem this work has proposed DermFusion, a novel dual-backbone CNN–Transformer framework that integrates ResNet-50+CBAM and ViT-B/16 via bidirectional cross-attention fusion for simultaneous skin lesion classification and segmentation, improving feature representation, prediction reliability, and confidence-aware clinical deployment.

III. Proposed Methodology

The CNN technique effectively extracts local texture and morphological lesion features but struggles with long-range dependencies, motivating ViTs and hybrid CNN–Transformer models that integrate local feature learning with global contextual understanding for improved classification performance. A bidirectional cross-attention fusion layer lets feature streams interact with each other so that each can pay attention to the other. This combines local and global information to improve image representation and analysis. An uncertainty-aware multi-task objective jointly learns multiple tasks, leveraging uncertainty estimation and deep supervision to enhance robustness. Post-hoc temperature scaling with MC dropout triage calibrates prediction confidence and estimates uncertainty, enabling reliable risk-aware decision-making and selective referral of uncertain cases for further review. In this study, we propose DermFusion, a hybrid dermatology framework that integrates four synergistic components: (i) a dual-backbone CNN–ViT architecture for complementary local-global feature extraction, (ii) a bidirectional cross-attention fusion mechanism for adaptive feature interaction, (iii) an uncertainty-aware multi-task optimization strategy with deep supervision, and (iv) a post-hoc calibration and uncertainty-guided triage module based on temperature scaling and Monte Carlo dropout (MC). An overview of the DermFusion framework, including its architectural design and workflow, is shown in Figure 1. Table II summarizes all mathematical notations and their definitions for reference throughout the subsequent discussion of the proposed model. The proposed DermFusion framework aims to maximize the expected log-joint probability of correct diagnoses and correct segmentation boundaries simultaneously, defined as Eq. (1).

$$\Theta^* = \arg\max_{\Theta} \ \mathbb{E}_{(X,Y,M)\sim\mathcal{D}}\big[\log P(Y, M \mid X, \Theta)\big] \quad (1)$$

where $X \in \mathbb{R}^{384\times384\times3}$ denotes an input image. Notation Y, M, and Θ denote the class label, segmentation mask, and collection of all trainable parameters, respectively. The complete optimization procedure implementing the objective is formalized, and each step and operation is described in the pseudocode of Algorithm 1.

TABLE II: Notations and their definition.

Symbol	Description
X	Input image $\in \mathbb{R}^{384\times384\times3}$
Y, M	Class label, segmentation mask
Θ	All trainable parameters
$F_{\text{cnn}}^{\text{raw}}$	Raw ResNet-50 C5 map
F_{cnn}	CBAM-refined CNN features $\in \mathbb{R}^{144\times2048}$
F_{vit}	ViT token sequence $\in \mathbb{R}^{576\times768}$
$\hat{F}_{\text{cnn}}, \hat{F}_{\text{vit}}$	Projected features $\in \mathbb{R}^{576\times768}$
$W_{\text{proj}}^{c}, W_{\text{proj}}^{v}$	Dimension projection matrices
W_{bottle}	Bottleneck projection $\mathbb{R}^{1536\times768}$
F_{fusion}	Fused representation $\in \mathbb{R}^{576\times1536}$
Q, K, V	Query, key, value
d, d_k	Shared dim (768); per-head dim (64)
h	Number of attention heads (12)
N_v	ViT token count OR Total pixel values per patch (576)
v_1, v_2	Learnable log-variance
L_{cls}	Focal classification loss
$L_{\text{seg,primary}}$	Focal-Tversky primary loss
L_{deep}	Deep supervision aggregate
$\lambda_1, \lambda_2, \lambda_3$	Deep supervision weights 0.4, 0.3, 0.3
TI	Tversky Index
$\alpha_{\text{FP}}, \beta_{\text{FN}}$	Tversky FP/FN penalties
$T_{\text{cal}}, T_{\text{cal}}^*$	Calibration temperature; optimal 1.75
T_{mc}	MC stochastic passes (10)
ECE	Expected Calibration Error
σ_{unc}^2	Scalar MC dropout variance
$H(y\|x)$	Predictive entropy
$\bar{p}_c$	Posterior mean for class c
τ, τ^*	Referral threshold; optimal 0.30
$U(\tau)$	Clinical utility score
C	Number of lesion classes (7)

A. Architectural Hypotheses

Prior to presenting the mathematical formulation of the proposed framework, we first introduce three design hypotheses that motivate the architectural design and component integration strategy of DermFusion. These hypotheses are empirically evaluated through the ablation study in Table XIV and Figure 7.

Hypothesis 1. *Feature Complementarity: CNN backbones encode high-frequency local texture through translation-invariant kernels, while ViT backbones encode global spatial dependencies through self-attention over the full token sequence. The bidirectional cross-attention fusion strategy improves class-discriminative representation learning by effectively combining complementary local and global features, a claim substantiated through empirical evaluation.*

Hypothesis 2. *Deep Supervision Gradient Flow: For multi-scale auxiliary supervision* $L_{\text{deep}} = \sum_{i=1}^{3} \lambda_i L_i^{\text{aux}}$ *with* $\lambda_i > 0$*, auxiliary gradients are routed directly to early decoder layers that would otherwise receive only the terminal task gradient. This mitigates gradient attenuation and improves optimization stability within the deep segmentation decoder, as demonstrated by the performance degradation observed when deep supervision is omitted.*

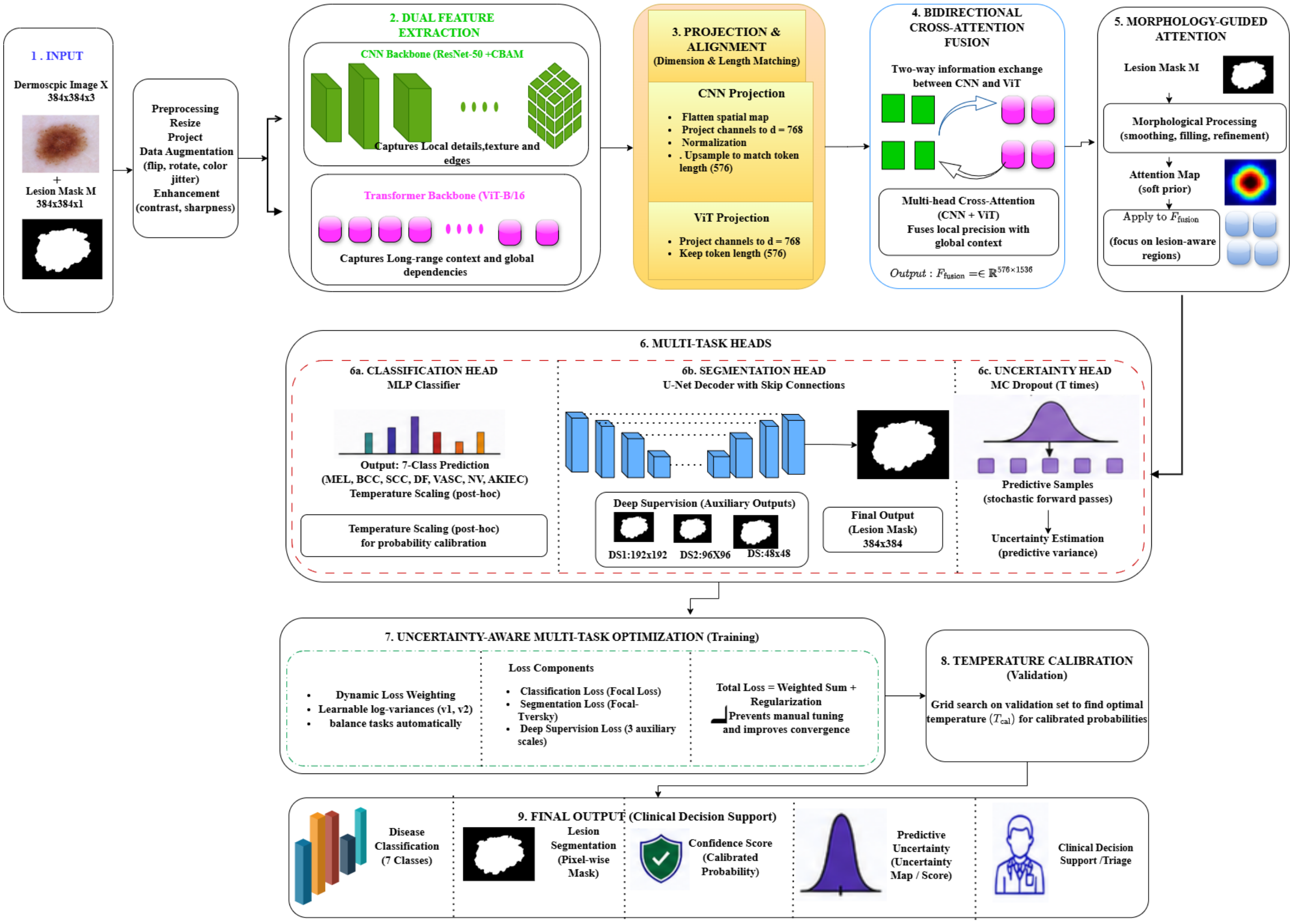

Fig. 1: Conceptual architecture and workflow of the proposed DermFusion model.

Hypothesis 3. *Bayesian Uncertainty Consistency: For MC stochastic predictions $\hat{p}_{t,c} \sim p(y = c|x, \Theta_t)$ with independently sampled dropout masks, the empirical predictive variance converges toward the true Bayesian posterior variance as $T_{\mathrm{mc}} \to \infty$ [21], providing a tractable uncertainty proxy for clinical referral validation.*

B. Feature Extraction

The proposed model used CNN-ViT as a backbone for feature extraction. The CNN backbone extracts hierarchical local representations through successive convolutional operations. It effectively captures texture, edge, color, and lesion-specific morphological patterns. In parallel, the ViT backbone partitions the image into patch embeddings and employs multi-head self-attention to model long-range spatial dependencies and global contextual relationships. These both take the same RGB image as input $X \in \mathbb{R}^{384\times384\times3}$.

1) CNN Backbone with Sequential CBAM Refinement: The CNN backbone ResNet-50 extracts hierarchical visual features through successive convolutional layers, progressively learning low-level edges, textures, and high-level semantic patterns. To enhance feature quality, a Sequential Convolutional Block Attention Module (CBAM) is applied after feature extraction. The proposed model seamlessly integrates channel-wise (what to look at) and spatial-wise (where to look) attention mechanisms into the intermediate layers of a deep convolutional network for feature refinement. CBAM sequential refinement enables the network to focus on salient lesion characteristics, improving feature discriminability and representation robustness. ResNet-50 initialized with ImageNet-pretrained weights is used for hierarchical local feature extraction. Given a 384×384 input image, the encoder follows a hierarchical feature extraction strategy, progressively reducing the spatial dimensions as $[384 \to 192 \to 96 \to 48 \to 24 \to 12]$ and expanding the channel depth as $[64 \to 256 \to 512 \to 1024 \to 2048]$. ResNet-50 processes the input image (X) through hierarchical residual blocks, producing deep semantic feature representations at stage (C_5), as defined in Eq. (2).

$$F^{\mathrm{raw}}_{\mathrm{cnn}} = f_{\mathrm{ResNet}}(X; \theta_{\mathrm{cnn}}) \in \mathbb{R}^{12\times12\times2048} \tag{2}$$

To suppress irrelevant background activations and enhance discriminative regions, CBAM [10] is sequentially applied using channel and spatial attention defined as Eq. (3) and (4).

$$F' = \sigma(M_c(F_{\text{cnn}}^{\text{raw}})) \odot F_{\text{cnn}}^{\text{raw}} \tag{3}$$

$$F_{\text{cnn}} = \sigma(M_s(F')) \odot F' \tag{4}$$

where $M_c \in \mathbb{R}^{1\times1\times2048}$ denotes channel attention and $M_s \in \mathbb{R}^{12\times12\times1}$ denotes spatial attention. After flattening, the spatial dimensions become $F_{\text{cnn}} \in \mathbb{R}^{144\times2048}$. The intermediate encoder stages ($C1$–$C5$) are additionally forwarded to the U-Net decoder through skip connections for multi-scale segmentation learning.

2) ViT Backbone: ViT treats an image as a sequence of patches, like tokens in NLP, which enables global context modeling through self-attention. ViT-B/16 partitions the 384×384 input image into non-overlapping 16×16 patches, $N_v = \left(\frac{384}{16}\right)^2 = 24^2 = 576$ visual tokens. Each patch is flattened and linearly projected into a 768-dimensional embedding, $F_{vit} \in \mathbb{R}^{576\times768}$, where 576 denotes the number of patches and 768 represents the token embedding dimension. Learnable positional embeddings are added to retain spatial information before transformer encoding. The ViT-B encoder consists of 12 transformer layers with 12 attention heads. The 768-dimensional embedding is equally divided across attention heads, yielding $d_k = \frac{768}{12} = 64$ dimensions per head. Each transformer block further includes a feed-forward network with a hidden dimension of $3072 = 4 \times 768$. Multi-head self-attention is formulated as $\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \ldots, \text{head}_h)W^O$, where head is mathematically expressed as Eq. (5).

$$\text{head}_i = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{5}$$

3) Dimension Projection for Cross-Attention: Dimension projection enables CNN-derived local texture and spatial features to interact effectively with Transformer-derived global contextual representations through cross-attention by mapping them into a common embedding space. Dimension projection for cross-attention compatibility ensures that features from different sources have the same embedding dimension before applying cross-attention. It enables efficient semantic alignment, stable optimization, and effective fusion of complementary local and global information extracted by different network branches. Since $F_{\text{cnn}} \in \mathbb{R}^{144\times2048}$ and $F_{\text{vit}} \in \mathbb{R}^{576\times768}$ have different spatial resolutions and feature dimensions, a two-stage projection strategy is used before cross-attention fusion. First, the CNN feature map is bilinearly upsampled from 12×12 to 24×24 to match the ViT token resolution. The upsampled feature map is then flattened into 576 spatial tokens using Eq. (6).

$$\tilde{F}_{\text{cnn}} = \text{flatten}\left(\text{Upsample}_{(2,2)}\left(F_{\text{cnn}}^{12\times12}\right)\right) \in \mathbb{R}^{576\times2048} \tag{6}$$

Then both CNN and ViT features are projected into a shared embedding space of dimension $d = 768$ using learnable linear projection layers defined in Eq. (7) and Eq. (8).

$$\hat{F}_{\text{cnn}} = \text{LayerNorm}\left(\tilde{F}_{\text{cnn}}W_{\text{proj}}^c\right), \quad W_{\text{proj}}^c \in \mathbb{R}^{2048\times768} \tag{7}$$

$$\hat{F}_{\text{vit}} = \text{LayerNorm}\left(F_{\text{vit}}W_{\text{proj}}^v\right), \quad W_{\text{proj}}^v \in \mathbb{R}^{768\times768} \tag{8}$$

This alignment produces compatible token representations for subsequent cross-attention fusion.

4) Bidirectional Cross-Attention Formulation: Bidirectional cross-attention provides a CNN branch to incorporate global contextual information and long-range dependencies captured by the Transformer and a Transformer branch to access fine-grained local textures, edge information, and spatial details preserved by the CNN. In our model, bidirectional cross-attention is used for reciprocal feature refinement between CNN and ViT branches. It allows local spatial details and global contextual information to be jointly exploited to enhance feature discriminability and improve image analysis performance. With both feature streams projected into $\mathbb{R}^{576\times768}$, bidirectional cross-attention is applied to model complementary interactions between CNN local features and ViT global representations. Let CNN features act as queries $Q_c = \hat{F}_{\text{cnn}}W_Q^c$,, and ViT features provide keys $K_v = \hat{F}_{\text{vit}}W_K^v$ and values $V_v = \hat{F}_{\text{vit}}W_V^v$ for attention which is expressed in Eq. (9).

$$A_c = \text{Softmax}\left(\frac{Q_cK_v^T}{\sqrt{d_k}}\right)V_v \tag{9}$$

Similarly, ViT features act as queries $Q_v = \hat{F}_{\text{vit}}W_Q^v$, and CNN features provide keys $K_c = \hat{F}_{\text{cnn}}W_K^c$ and values $V_c = \hat{F}_{\text{cnn}}W_V^c$ for attention, which is expressed in Eq. (10).

$$A_v = \text{Softmax}\left(\frac{Q_vK_c^T}{\sqrt{d_k}}\right)V_c \tag{10}$$

where $d_k = 64$ denotes the feature dimension per attention head with $h = 12$ heads. The multi-head cross-attention (MHCA) is formulated in Eq. (11).

$$\text{MHCA}(Q, K, V) = \text{Concat}\left(\text{head}_1, \ldots, \text{head}_h\right)W^O \tag{11}$$

The outputs from both attention directions are concatenated and layer-normalized to produce the fused representation using Eq. (12).

$$F_{\text{fusion}} = \text{LayerNorm}\left(\text{Concat}(A_c, A_v)\right) \in \mathbb{R}^{576\times1536} \tag{12}$$

Finally, a learnable bottleneck projection $W_{\text{bottle}} \in \mathbb{R}^{1536\times768}$ compresses the fused features into $\mathbb{R}^{576\times768}$ before forwarding them to the downstream prediction heads.

C. Uncertainty-Aware Multi-Task Optimization

For simultaneously classification and segmentation using CNN-ViT architectures, uncertainty-aware multi-task optimization dynamically adjust the contribution of each during training according to its estimated uncertainty. Instead of assigning fixed weights to task losses, the model learns which tasks are more reliable and adjusts their influence automatically. It jointly optimizes four loss functions governing classification, segmentation, calibration, and uncertainty estimation to promote balanced learning and enhance the reliability of the proposed DermFusion framework. The homoscedastic loss balancing, focal classification loss, segmentation decoder with deep supervision, post-hoc temperature scaling, and MC dropout uncertainty quantification components collectively implement an uncertainty-aware multi-task optimization framework, where each module addresses a specific aspect of learning reliability, task balancing, and prediction confidence.

1) Homoscedastic Loss Balancing: It is the core optimization mechanism that automatically weights classification, segmentation, calibration, and uncertainty objectives according to their learned task uncertainties, preventing manual loss tuning. To avoid manual tuning of multi-task loss weights, homoscedastic uncertainty weighting [22] is adopted. It is expressed in Eq. (13).

$$L_{\text{total}} = e^{-v_1} L_{\text{cls}} + e^{v_2} L_{\text{seg}}^{\text{total}} + (v_1 + v_2) \tag{13}$$

where $L_{\text{seg}}^{\text{total}} = L_{\text{seg,primary}} + L_{\text{deep}}$ and v_1, v_2 are learnable log-variance parameters initialized to zero.

2) Focal Classification Loss: In multi-task models that include a classification objective, conventional cross-entropy loss often struggles to learn effectively from highly imbalanced datasets. The massive volume of "easy" negative examples overwhelms the gradient, causing the model to neglect the rare, harder-to-classify cases. The focal classification loss is a specialized loss function designed to handle severe class imbalances during classification and emphasizes on minority-class samples by down-weighting to improving class discrimination. The classification branch minimizes focal loss [26] using the Eq. (14).

$$L_{\text{cls}} = -\sum_i \alpha_i^{(\text{cls})} (1 - \hat{p}_i)^{\gamma_f} \log(\hat{p}_i), \quad \gamma_f = 2 \tag{14}$$

where $\alpha_i^{(\text{cls})}$ denotes class-balancing weights inversely proportional to class frequency.

3) Segmentation Decoder with Deep Supervision: Segmentation decoder with deep supervision generates lesion masks from fused CNN–ViT features and applies auxiliary segmentation losses at intermediate decoder stages. Deep supervision solves the vanishing gradient problem by forcing intermediate stages of the decoder to output their own "mini" segmentation predictions. This strengthens gradient propagation, improves optimization stability, and enhances boundary-aware segmentation performance within the uncertainty-aware multi-task learning framework. A U-Net decoder is an encoder–decoder segmentation module that progressively upsamples deep features and combines them with encoder skip connections to generate accurate high-resolution pixel-wise segmentation masks. The U-Net decoder receives bottleneck-reduced features together with multi-scale skip connections from all ResNet stages (C1-C5). Deep supervision is applied using three auxiliary segmentation losses. These are defined in Eq. (15).

$$L_{\text{deep}} = \sum_{i=1}^{3} \lambda_i L_i^{\text{aux}} \tag{15}$$

where $\lambda_1 = 0.4, \lambda_2 = 0.3, \lambda_3 = 0.3$ The segmentation branch employs the Focal-Tversky index [27] as a similarity measure used in image segmentation and defined in Eq. (16). It provides a class-imbalance-aware overlap measure that enables explicit control over false-positive and false-negative penalties and generalizes both the Dice coefficient and Jaccard index through different penalties for false positives (FP) and false negatives (FN).

$$TI = \frac{TP}{TP + \alpha_{\text{FP}} FP + \beta_{\text{FN}} FN}, \tag{16}$$

where $\alpha_{\text{FP}} = 0.3$ and $\beta_{\text{FN}} = 0.7$ The final segmentation objective combines binary cross-entropy and focal Tversky loss, as these two losses address different aspects of the segmentation problem and complement each other during optimization. The combined loss is defined in Eq. (17).

$$L_{\text{seg,primary}} = 0.4 L_{\text{BCE}} + 0.6(1 - TI)^{\gamma_s}, \tag{17}$$

where $\gamma_s = 1.33$. The BCE, Tversky, and focusing parameters ensure pixel-level supervision, optimization of region overlap, and concentrated optimization on hard-to-segment pixels, respectively.

4) Post-Hoc Temperature Scaling: It is not necessary that if the prediction scores of a model are high, they always accurately represent the model's reliability. Therefore, in this work, after the model gets fully trained, a calibration technique, post-hoc temperature scaling, is applied to assess the model's actual predicted probabilities better to reflect the true probability of being correct [7]. It is defined in Eq. (18).

$$\hat{p}_i = \frac{\exp(z_i / T_{\text{cal}})}{\sum_j \exp(z_j / T_{\text{cal}})} \tag{18}$$

The optimal calibration temperature is obtained by minimizing Expected Calibration Error (ECE), which is expressed in Eq. (19).

$$T_{\text{cal}}^* = \arg\min_{T_{\text{cal}} > 0} ECE\left(\hat{p}(T_{\text{cal}}), y_{\text{val}}\right) \tag{19}$$

5) MC Dropout Uncertainty Quantification: To estimate how confident a model is in its predictions instead of producing only a class label or segmentation mask, we have used MC dropout uncertainty quantification for a measure of the proposed model's predictive uncertainty. At inference time, $T_{\text{mc}} = 10$ stochastic forward passes with dropout enabled approximate Bayesian predictive uncertainty, as expressed in Eq. (20).

$$\bar{p}_c(y = c|x) \approx \frac{1}{T_{\text{mc}}} \sum_{t=1}^{T_{\text{mc}}} p_{t,c}(y = c|x) \tag{20}$$

Predictive variance across all classes is computed as using Eq. (21).

$$\sigma_{\text{unc}}^2 = \frac{1}{C} \sum_{c=1}^{C} \left[\frac{1}{T_{\text{mc}}} \sum_{t=1}^{T_{\text{mc}}} \hat{p}_{t,c}^2 - \bar{p}_c^2 \right] \tag{21}$$

The estimate the predictive entropy is additionally computed defined Eq. (22).

$$H(y|x) = -\sum_{c=1}^{C} \bar{p}_c \log \bar{p}_c \tag{22}$$

Calibration quality is evaluated using ECE employing Eq. (23).

$$ECE = \sum_{m=1}^{M} \frac{|B_m|}{n} \left| acc(B_m) - conf(B_m) \right| \tag{23}$$

where B_m denotes the set of samples in confidence bin m, $acc(B_m)$ is the accuracy of bin m, and $conf(B_m)$ is the mean confidence of the bin.

The following three algorithms provide complete pseudocode that enables independent re-implementation by clearly specifying all procedures, computational steps, and methodological details. Algorithm 1 details the complete training procedure, incorporating multi-task loss optimization and post-training temperature calibration for reliability enhancement. Algorithm 2 describes the bidirectional cross-attention mechanism integrated into the training process. Algorithm 3 handles inference-time uncertainty triage using the optimal threshold τ^* determined through the validation sweep reported in Table IV.

Algorithm 1 Multi-stage optimization of DermFusion.

Require: Data $\mathcal{D} = \{(x_i, y_i, m_i)\}_{i=1}^{N}$ (Table III),lr $\eta = 10^{-4}$, epochs $E = 45$, freeze epochs $E_f = 6$
Ensure: Θ^*, T^*_{cal}

Initialise ResNet-50 and ViT-B/16 from ImageNet weights
Initialise projection matrices $W^c_{\text{proj}}, W^v_{\text{proj}}$ (Eqs. (7)-(8)), cross-attention heads (Eqs. (9)-(10)), and decoder heads with Xavier uniform
Set $v_1 \leftarrow 0$, $v_2 \leftarrow 0$
Freeze backbone parameters for $E_f = 6$ warmup epochs
for epoch $= 1$ **to** E **do**
    Sample augmented mini-batch (x, y, m) from $\mathcal{D}$
    if epoch $> E_f$ **then** Unfreeze all backbone parameters
    end if
    $\hat{F}_{\text{cnn}} \leftarrow \text{LayerNorm}\big(\text{Up}_2(\text{CBAM}(f_{\text{R}}(x)))\, W^c_{\text{proj}}\big)$
    $\hat{F}_{\text{vit}} \leftarrow \text{LayerNorm}(f_{\text{ViT}}(x)\, W^v_{\text{proj}})$
    $F_{\text{fusion}} \leftarrow$ Algorithm $2(\hat{F}_{\text{cnn}}, \hat{F}_{\text{vit}})$
    $F_{\text{bt}} \leftarrow F_{\text{fusion}} W_{\text{bottle}} \in \mathbb{R}^{576\times 768}$
    Compute $\hat{y}$ (classification head) and $\hat{m}$ (U-Net decoder)
    $L_{\text{total}} \leftarrow e^{-v_1} L_{\text{cls}} + e^{-v_2}(L_{\text{seg,primary}} + L_{\text{deep}}) + (v_1 + v_2)$
    Clip gradient norm ≤ 1.0; update Θ via AdamW.
end for
Optimize T^*_{cal} via grid search on validation set using Eq. (19).
return Θ^*, T^*_{cal}

Algorithm 2 Bidirectional CNN-Transformer cross-attention fusion.

Require: $\hat{F}_{\text{cnn}} \in \mathbb{R}^{576\times 768}$ $\hat{F}_{\text{vit}} \in \mathbb{R}^{576\times 768}$
Ensure: $F_{\text{fusion}} \in \mathbb{R}^{576\times 1536}$

$Q_c, K_v, V_v \leftarrow \hat{F}_{\text{cnn}} W^c_Q,\ \hat{F}_{\text{vit}} W^v_K,\ \hat{F}_{\text{vit}} W^v_V$
$A_c \leftarrow \text{MHCA}(Q_c, K_v, V_v)$ ▷ Eq. (9), $h = 12$ heads, $d_k = 64$
$Q_v, K_c, V_c \leftarrow \hat{F}_{\text{vit}} W^v_Q,\ \hat{F}_{\text{cnn}} W^c_K,\ \hat{F}_{\text{cnn}} W^c_V$
$A_v \leftarrow \text{MHCA}(Q_v, K_c, V_c)$
$F_{\text{fusion}} \leftarrow \text{LayerNorm}(\text{Concat}(A_c, A_v))$
return F_{fusion}

Algorithm 3 MC Bayesian uncertainty estimation and clinical referral.

Require: Test sample x, $T_{\text{mc}} = 10$, threshold τ^* from Table IV, T^*_{cal} from Eq. (19)
Ensure: Prediction $\bar{p}$, scalar variance σ^2_{unc}, and clinical decision

for $t = 1$ **to** T_{mc} **do**
    Activate classifier-head dropout (rate 0.5) and cross-attention projection dropout (rate 0.1).
    $p_{t,c} \leftarrow \text{Softmax}\big(f_\Theta(x; \text{dropout})/T^*_{\text{cal}}\big)_c$
end for
$\bar{p}_c \leftarrow \frac{1}{T_{\text{mc}}} \sum_t p_{t,c}$
$\sigma^2_{\text{unc}} \leftarrow \frac{1}{C} \sum_c \big[\frac{1}{T_{\text{mc}}} \sum_t p^2_{t,c} - \bar{p}^2_c\big]$
if $\sigma^2_{\text{unc}} > \tau^*$ **then**
    Refer to human expert
else
    Accept autonomous prediction $\hat{y} = \arg\max_c \bar{p}_c$
end if
return $\bar{p}$, σ^2_{unc}, decision

IV. Dataset Description and Experimental Setup

This section describes the datasets, training configuration, baseline definitions, threshold selection procedure, and statistical testing protocols used to evaluate DermFusion. Reproducibility is maintained by fixing random seeds across experiments and publicly providing patient-wise data split indices.

A. Primary Dataset

In this work we have used the HAM10000 dataset [9], which is a publicly available multi-source dermoscopy benchmark containing 10,015 images spanning seven clinically relevant categories: melanoma (MEL), melanocytic nevi (NV), basal cell carcinoma (BCC), actinic keratosis / intraepithelial carcinoma (AKIEC), benign keratosis (BKL), dermatofibroma (DF), and vascular lesions (VASC). To prevent. To avoid patient identity leakage, where images from the same lesion could appear in multiple partitions, a strict patient-wise split is enforced using unique $lesion\_id$ metadata, ensuring complete separation between training, validation, and test sets. Table III summarizes the resulting class-stratified partition: 6,009 training samples (folds 2-4), 2,003 validation samples (fold 1), and 2,003 test samples (fold 0), with all random seeds fixed to 42. The NV class constitutes 66.9% of total samples, creating a 58:1 class imbalance ratio with DF that is addressed via focal loss (Eq. (14)) and per class weight $\alpha_i^{(\text{cls})}$. The observed class imbalance necessitates the use of class-frequency-weighted

TABLE III: Patient-wise stratified distribution of the HAM10000 dataset across training, validation, and test sets.

Class	Total	Train	Val	Test
MEL	1113	667	223	223
NV	6705	4023	1341	1341
BCC	514	308	103	103
AKIEC	327	196	65	66
BKL	1099	659	220	220
DF	115	69	23	23
VASC	142	85	28	29
Total	**10015**	**6009**	**2003**	**2003**

focal loss, with its impact evident in the class-wise precision results presented in Table VII.

B. Segmentation Annotation Source

Ground-truth segmentation masks come from the ISIC 2018 Task 1 lesion boundary annotation set [13], which provides pixel-level binary masks for each lesion. Masks were paired with HAM10000 classification images using the unique $lesion\_id$ metadata field. These segmentation masks act as the target variable M in the joint optimization objective defined in Eq. (1) and facilitate computation of the Focal-Tversky loss functions defined in Eqs. (16)–(17). Segmentation results are reported in Table VI.

C. Training Configuration

Model training ran on a single NVIDIA Tesla T4 GPU with 15.6 GB VRAM, employing the AdamW optimizer, cosine annealing scheduling ($\eta_0 = 10^{-4}$), and FP16 mixed-precision computation. The effective batch size of 16 has been achieved through 4×4 gradient accumulation. The model contains 124.3M parameters in total, with only 14.4M trained during the 6-epoch backbone warmup phase of Algorithm 1. Subsequently, all parameters were unfrozen and optimized throughout the remaining 39 training epochs. Tversky hyperparameters ($\alpha_{\mathrm{FP}} = 0.3$, $\beta_{\mathrm{FN}} = 0.7$, $\gamma_s = 1.33$) and deep supervision weights ($\lambda_1 = 0.4$, $\lambda_2 = 0.3$, $\lambda_3 = 0.3$) were determined through an extensive grid-search procedure conducted on the validation dataset. Augmentation comprised random rotation ($p = 0.5$), horizontal flip ($p = 0.5$), colour jittering, and MixUp ($p = 0.3$). Table XV presents the comparative computational complexity and resource requirements of the proposed model and baselines.

D. Baseline Definitions

For the calibration comparison, the CNN+ViT Fusion baseline is a simplified version of the proposed architecture. Instead of using the bidirectional cross-attention mechanism, it directly concatenates the 2048-dimensional CNN global pooled features with the 768-dimensional ViT [CLS] token ($\in\mathbb{R}^{768}$) and passes them to a linear classifier. This baseline excludes the segmentation branch, deep-supervision strategy, and MC-dropout uncertainty estimation, thereby isolating the effect of advanced fusion and uncertainty-aware components. Temperature scaling with the optimal calibration temperature $T^*_{\mathrm{cal}} = 1.75$ has been uniformly applied across all four compared methods and results shown in Table XVI.

E. Uncertainty Threshold Selection

The referral threshold τ in Algorithm 3 governs the trade-off between prediction coverage and classification accuracy reported in Table IV. Specifically, τ was swept across seven candidate values $\tau \in \{0.10, 0.15, 0.20, 0.25, 0.30, 0.35, 0.40\}$, and the optimal value was selected by maximizing a clinical utility score $U(\tau)$, as stated in Eq. (24) subject to the clinical safety constraint $\mathrm{MEL\,Sens_{ret}} \geq 95\%$ on the validation set.

$$U(\tau) = 0.7 \cdot \mathrm{Acc_{ret}}(\tau) - 0.3 \cdot \big(\mathrm{Coverage}(\tau)\big) \tag{24}$$

The coefficients 0.7 and 0.3 in Eq. (24) reflect the clinical priority of accuracy in autonomous cases relative to the penalty for excessive referral rates. These represent one clinically credible operating point and should be revisited by deploying clinicians. As shown in Table IV, the optimal threshold $\tau^* = 0.30$ maximizes the utility score $U = 0.702$ and was fixed entirely on the validation set before any test-set evaluation. It is clear from Table IV that as τ decreases, coverage increases, but retained accuracy falls. The utility function $U(\tau)$ balances these competing objectives. The optimal value, $\tau^* = 0.30$, is highlighted in bold and was fixed before any test set access.

TABLE IV: Sensitivity analysis of referral thresholds on the validation dataset.

τ	**Coverage(%)**	**Acc Ret.(%)**	**MEL Sens(%)**	$U(\tau)$
0.10	94.2	88.5	86.1	0.591
0.15	89.6	90.1	89.4	0.601
0.20	83.1	91.8	93.0	0.641
0.25	76.4	93.5	95.2	0.666
0.30*	**68.0**	**96.4**	**97.1**	**0.702**
0.35	61.2	97.0	97.8	0.698
0.40	53.5	97.8	98.1	0.679

F. Statistical Testing

The statistical robustness of the proposed framework was evaluated through three complementary levels. First, a 5-fold cross-validation stability analysis (Table XII) was conducted, where DeLong's test [28] compared paired AUC estimates across folds. The resulting p-values $p > 0.05$ for all fold pairs, indicating no statistically significant variability and demonstrating consistent model performance. The threshold selection procedure is fully described in Section IV-E; briefly, $\tau^* = 0.30$ was fixed on the validation set before any test-set access. Second, for SOTA comparisons in Tables IX and X, overlapping bootstrapped 95% CIs (1000 resamples) are used as evidence of non-separable performance ranges, since baselines used different dataset splits, precluding paired testing; all SOTA comparisons are explicitly labeled *indicative*. Third, for the demographic fairness analysis in Table XVII, subgroup accuracy differences were evaluated using a two-proportion z-test with Bonferroni correction across four comparisons (corrected threshold $\alpha_{\mathrm{adj}} = 0.0125$). All 95% confidence intervals were computed via non-parametric bootstrapping with 1000 resamples.

V. Results, Clinical Analysis and Findings

This section summarizes the quantitative evaluation of the proposed framework on the HAM10000 test partition and external validation datasets. Results are presented systematically, beginning with classification and segmentation performance, followed by per-class diagnostic analyses. Subsequent sections compare the proposed method with state-of-the-art approaches, assess external robustness evaluation generalization capability, examine cross-validation stability, evaluate uncertainty-based referral performance, analyze the contribution of individual components through ablation studies, and finally report computational complexity and resource requirements.

A. Classification Performance

Table V summarizes the overall classification performance achieved by the proposed model on the held-out test fold, comprising 2,003 samples. All reported 95% confidence intervals were estimated using 1000 bootstrap resamples, as described in Section IV-F. MEL-Adj. denotes the melanoma sensitivity-adjusted decision threshold (≥ 0.45) applied during model evaluation. ECE is calculated using Eq. (23) with a bin count of $M = 15$. At the default decision threshold, DermFusion attains an accuracy of 87.82% and a ROC-AUC of 0.9799, outperforming the strongest competing baseline, which achieved 87.2% accuracy and 0.975 ROC-AUC. With a sensitivity-adjusted melanoma decision boundary (≥ 0.45, selected on the validation set where it gave MEL sensitivity of 90.8% at 86.1% overall accuracy), melanoma sensitivity on the *test fold* rises from 88.80% to 91.2% at the cost of only 1.22% overall accuracy, with all other metrics remaining within their bootstrap confidence intervals. Temperature scaling reduced the ECE from 0.4395 to 0.1575, indicating substantially improved probability calibration, as illustrated in Figure 2 through a comparison of calibrated and uncalibrated model predictions.

TABLE V: Performance analysis of the proposed model on the HAM10000 test fold.

Metric	Default	95% CI	MEL-Adj.
Accuracy	87.82%	[86.9, 88.6]	86.60%
Balanced Accuracy	87.77%	[87.0, 88.5]	87.80%
Macro F1-score	0.8497	[0.838, 0.861]	0.8450
ROC-AUC	0.9799	[0.974, 0.985]	0.9799
PR-AUC (macro)	0.9172	[0.912, 0.922]	0.9172
MEL Sensitivity	88.80%	[86.5, 90.9]	**91.20%**
ECE (Eq. (23))	0.1575	[0.149, 0.165]	–

Figure 2 displays reliability diagrams that correspond to Table V, comparing calibration performance through confidence–accuracy alignment across evaluated methods. The uncalibrated model (red, ECE= 0.4395 computed using Eq. (23)) exhibits substantial overconfidence consistently across all confidence bins. Temperature scaling with an optimal

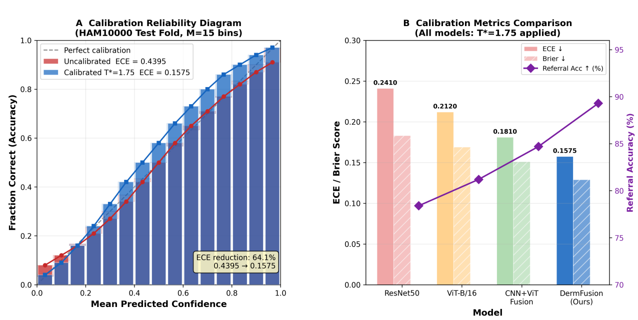

Fig. 2: Reliability diagrams comparing calibration performance across methods in Table V.

calibration temperature of $T^*_{\text{cal}} = 1.75$ (Eq. (18), blue curve) substantially improves confidence calibration, restoring near-diagonal reliability and reducing ECE from 0.4395 to 0.1575, corresponding to a 64.1% relative improvement. The right side presents a comparative evaluation of ECE and Brier scores for RestNet 50, ViT-B/16, CNN+ViT, and DermFusion methods, as also summarized in Table XVI.

B. Segmentation Performance

Lesion boundary segmentation is the second output task of DermFusion and is jointly optimized with classification through the shared cross-attention representation (Eq. (12)) and the U-Net decoder. Two standard overlap metrics have been used. The Dice Similarity Coefficient (DSC) is defined as $\text{DSC} = 2\,|P \cap G|\,/\,(|P| + |G|)$, where P is the predicted binary mask and G is the ground-truth mask; it equals 1.0 for perfect overlap and 0.0 for complete mismatch. The Intersection over Union (IoU), also called the Jaccard Index, is defined as $\text{IoU} = |P \cap G|\,/\,|P \cup G|$ and is always lower than the DSC for the same prediction. Both metrics are computed at the primary decoder output resolution (384×384) using the primary Focal-Tversky loss from Eq. (17). The Dice score is computed at the primary output resolution (384×384) using the primary loss in Eq. (17), and 95% CIs are bootstrapped (1000 resamples).

TABLE VI: Segmentation performance on the HAM10000 test fold.

Metric	Value	95% CI
Dice Score	0.9359	[0.929, 0.941]
IoU Score	0.8880	[0.881, 0.895]

Table VI presents these metrics on the 2,003-sample test fold. DermFusion achieves strong segmentation performance with a Dice score of 0.9359 (95% CI: 0.929–0.941) and IoU of 0.8880 (95% CI: 0.881–0.895). These results are noteworthy for two reasons. First, DermFusion exceeds every baseline listed in Table X that uses only segmentation, including Swin-Unet (Dice = 0.918, IoU = 0.851, using 108M parameters), FAT-Net (Dice = 0.910), and SegFormer-B3 (Dice = 0.911), while also optimizing a seven-class classification head. Second, the Focal-Tversky formulation in Eq. (16) sets $\beta_{\text{FN}} = 0.7$ to penalize missed lesion pixels more heavily than over-segmentation. This clinically motivated asymmetry shows up in the narrow confidence intervals, indicating consistent boundary delineation across the test fold. The multi-scale deep supervision in Eq. (15) improves performance by +0.0217 Dice compared to the version without it, as shown in Table XIV.

C. Per-Class Evaluation and Confusion Analysis

Although overall accuracy reflects general model performance, per-class analysis identifies specific lesion categories responsible for errors and provides insight into the underlying reasons for these misclassifications and performance variations. The results for per class are analyzed using four perspectives: precision, recall, F1 score (Table VII and Figure 3), a full confusion matrix (Table VIII), and per-class ROC curves (Figure 4), providing a comprehensive evaluation of classification performance across different lesion categories and visualization methods. Precision, recall, and F1 score are derived from the confusion matrix shown in Table VIII. The row totals

(MEL 205, NV 1361, BCC 93, AKIEC 67, BKL 223, DF 22, VASC 32) exactly correspond to the class sample counts n, confirming consistency between the confusion matrix and dataset distribution. Low MEL precision reflects deliberate sensitivity prioritization via $\alpha_i^{(\mathrm{cls})}$ in Eq. (14).

The results shown in Table VII reveal reduced precision for MEL (0.548), reflecting the model's deliberate emphasis on sensitivity. By assigning greater importance to melanoma detection through $\alpha_{\mathrm{MEL}}^{(\mathrm{cls})}$ in Eq. (14), the model tolerates additional false positives to reduce the likelihood of missed melanoma diagnoses. This results in increased false-positive melanoma predictions, a clinically preferred trade-off for reducing the likelihood of missed melanomas. Recall (sensitivity) measures how many true instances of that class are detected. MEL recall of 0.888 means 88.8% of true melanomas are caught at the default threshold, rising to 91.2% with the adjusted threshold (Table V). The F1 score is the harmonic mean of precision and recall and provides a balanced summary. VASC achieves the highest F1 (0.970) owing to its visually distinctive capillary morphology, while MEL achieves the lowest (0.678), reflecting the precision–recall trade-off induced by the clinical weighting $\alpha_{\mathrm{MEL}}^{(\mathrm{cls})}$ in Eq. (14). Despite the severe NV-to-DF imbalance of 58:1 reported in Table III, the DF class attains a recall of 0.955, highlighting the effectiveness of the inverse-frequency weighting strategy in Eq. (14).

TABLE VII: The diagnostic performance for each class assessment on the HAM10000 test fold.

Class	Precision	Recall	F1	n
MEL	0.548	0.888	0.678	205
NV	0.969	0.899	0.933	1361
BCC	0.829	0.989	0.902	93
AKIEC	0.780	0.687	0.730	67
BKL	0.895	0.726	0.802	223
DF	0.913	0.955	0.933	22
VASC	0.941	1.000	0.970	32

Figure 3 visualizes the results shown in Table VII across two panels. Panel (A) plots the three scores side-by-side for each class. Panel (B) plots sensitivity (recall) and specificity with the test set sample count annotated per class. All seven classes maintain sensitivity $> 85\%$ and specificity $> 96\%$, confirming that no class systematically dominates errors. In Figure 3, panel (A) shows the precision, recall, and F1 score per lesion type. It is clear from the results shown that despite the 58:1 NV:DF imbalance, robust recall is maintained via focal weighting. MEL recall is 0.888 at the default threshold and rises to 0.912 with the adjusted threshold (Table V). Panel (B) shows the sensitivity and specificity per class with test-set counts from Table III. It is clear that all seven classes maintain sensitivity $> 85\%$ and specificity $> 96\%$, confirming that no class systematically dominates errors.

The confusion matrix obtained from the HAM10000 test fold ($n = 2003$) is shown in Table VIII, where the row sums correspond to the per-class test counts listed in Table VII. Diagonal entries are correct classifications. The clinically critical NV→MEL error (38 cases) is substantially lower than the NV→BKL error (73 cases), because both BKL confusions involve clinically benign classes. The cross-attention

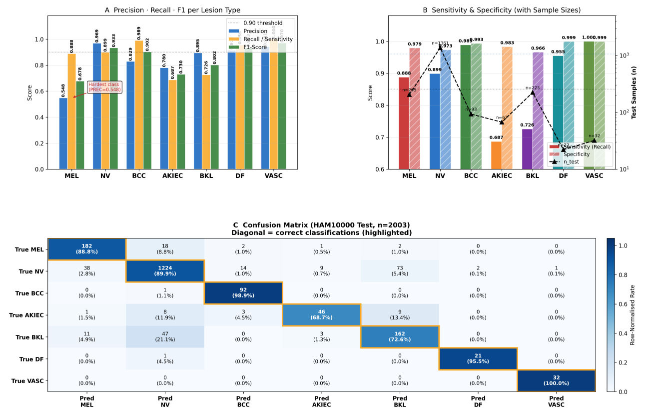

Fig. 3: Per-class performance on the HAM10000 test fold, detailed in table VII.

in Eqs. (9)-(10) and the sensitivity weighting in Eq. (14) are designed to minimize MEL miss rates. In Table VIII, each entry (i, j) represents the number of samples from the true class i predicted as class j, with dominant diagonal values indicating strong classification performance. Off-diagonal entries reveal specific confusion patterns. The predominant off-diagonal entry corresponds to NV→MEL, indicating 38 nevus lesions falsely identified as melanoma. This is the safest type of error in clinical practice, a false alarm that triggers a biopsy rather than the reverse (MEL→NV, 18 cases), which represents a missed melanoma. The bidirectional cross-attention (Eqs. (9)-(10)) reduces these confusions by enabling the ViT branch to integrate global morphological context that distinguishes NV from MEL. The NV→BKL confusion (73 cases) is the second largest off-diagonal entry. Both are benign, so the confusion has lower clinical impact. VASC and BCC are almost perfectly separated, with 32/32 and 92/93 correct classifications, respectively.

TABLE VIII: Confusion matrix from the HAM10000 test fold ($n = 2003$).

		Predicted						
		MEL	NV	BCC	AKIEC	BKL	DF	VASC
True	MEL	182	18	2	1	2	0	0
	NV	38	1224	14	9	73	2	1
	BCC	0	1	92	0	0	0	0
	AKIEC	1	8	3	46	9	0	0
	BKL	11	47	0	3	162	0	0
	DF	0	1	0	0	0	21	0
	VASC	0	0	0	0	0	0	32

The Receiver Operating Characteristic (ROC) curve for each class is generated using a one-versus-rest protocol at the calibrated probabilities $\hat{p}_i$ from Eq. (18) ($T^*_{\mathrm{cal}} = 1.75$). Per-class ROC curves shown in Figure 4 are obtained from the HAM10000 test, $T^*_{\mathrm{cal}} = 1.75$ from Eq. (18), and one-versus-rest classification. Each panel corresponds to one class in Table VII. The macro-average panel aggregates all seven. A higher Area Under the Curve (AUC) indicates better class separability independent of any fixed threshold. As shown in Figure, ROC, DermFusion achieves strong per-class AUC values: MEL 0.984, NV 0.988, BCC 0.970, AKIEC 0.955, BKL 0.961, DF 0.948, and VASC 0.972, yielding a macro-average of 0.9799. The DF staircase artifact in the 0.5–0.8

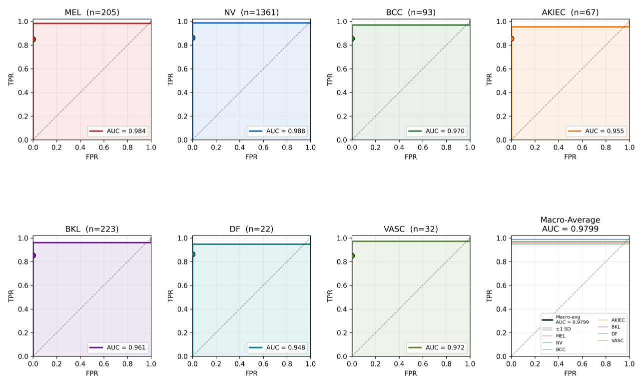

Fig. 4: Per-class ROC curves obtained from the HAM10000 test, $T^*_{\text{cal}} = 1.75$ from one-versus-rest.

FPR range is due to the small $n = 22$ test partition and is not a classifier anomaly. Each curve traces the trade-off between sensitivity (true positive rate) and $1-$specificity (false positive rate) as the decision threshold varies continuously from 0 to 1.

D. State-of-the-Art Benchmarking

Performance comparison of different baseline classification models trained on the HAM10000 dataset and their results shown in Tables IX and X. These different baseline classification models were reproduced from the original publication, which used different dataset splits and preprocessing. Table IX shows that DermFusion is the only method evaluated using a split-controlled protocol (Ctrl.=Yes), where predictions are compared under controlled conditions. The reported accuracy and AUC values are point estimates, while their statistical reliability is provided through bootstrapped 95% confidence intervals in Table V. All baseline results are taken from the respective original studies. These comparisons are indicative only. The Ctrl. column explicitly marks results obtained under the patient-wise split of Section IV-A. Bootstrapped 95% CIs (Section IV-F) are used to characterize performance ranges. The DeLong's paired AUC test [28] cannot be applied to baselines evaluated on different test sets.

TABLE IX: Performance comparison of different baseline classification models trained on the HAM10000 dataset.

Method	Year	Acc(%)	AUC	F1	Params(M)	Ctrl.
ResNet50 [2]	2016	81.3	0.921	0.771	25.6	No
DenseNet121 [3]	2017	83.4	0.944	0.792	8.0	No
EfficientNet-B4 [4]	2019	85.7	0.961	0.814	19.3	No
ConvNeXtV2 [12]	2023	86.8	0.965	0.829	88.5	No
Hybrid MedViT	2023	87.2	0.975	0.842	112.0	No
MedMamba [19]	2024	86.5	0.969	0.831	28.4	No
DermFusion	2026	**87.82**	**0.9799**	**0.8497**	**124.3**	**Yes**

Table X compares segmentation performance on skin lesion benchmarks. DermFusion is the only multitask model evaluated under a split-controlled protocol. Since baseline Dice and IoU scores have been reported using different splits in the original studies, direct numerical comparisons should be interpreted as indicative rather than strictly equivalent. Table X shows that DermFusion's classification accuracy (87.82%) and AUC (0.9799) exceed all compared methods, including the next-best Hybrid MedViT (87.2%, 0.975), under the only split-controlled evaluation. Table X shows that DermFusion's Dice (0.9359) and IoU (0.8880) exceed all compared segmentation methods, including the strongest pure-segmentation baseline, Swin-Unet (0.918, 0.851).

TABLE X: Benchmark segmentation analysis of DermFusion and prior methods on skin lesions.

Method	Year	Dice	IoU	Task	Params(M)	Ctrl.
UNet++ [20]	2018	0.892	0.811	Seg	36.6	No
TransUNet [15]	2021	0.903	0.829	Multi-task	105.0	No
Swin-Unet [16]	2022	0.918	0.851	Seg	108.0	No
FAT-Net [18]	2022	0.910	0.842	Seg	28.14	No
SegFormer-B3 [17]	2022	0.911	0.844	Seg	47.3	No
DermFusion	2026	**0.9359**	**0.8880**	**Multi-task**	**124.3**	**Yes**

E. external robustness evaluation Multi-Dataset Validation

A model trained on a single institution's dermoscopy pipeline may fail when deployed with different acquisition equipment, imaging protocols, or patient demographics. To measure this domain generalization capability without any adaptation, DermFusion trained exclusively on HAM10000 (Table III) was evaluated on three other independent external datasets with no fine-tuning, no preprocessing modification, and temperature scaling fixed at $T^*_{\text{cal}} = 1.75$. The three datasets PH^2, ISIC 2019, and PAD-UFES-20 differ from HAM10000, which has been taken for external robustness evaluate multi-dataset validation in this work. External robustness evaluation generalized across three PH^2, ISIC 2019, and PAD-UFES-20 independent datasets with no fine-tuning. The model has trained only on HAM10000 (Table III) and applied directly. Temperature scaling is fixed at $T^*_{\text{cal}} = 1.75$ (Eq. (19)). AUC 95% CIs are bootstrapped with 1000 resamples (Section IV-F). Performance degrades with increasing domain shift: PH^2 (clinical dermoscope, European) $>$ ISIC 2019 (multi-center) $>$ PAD-UFES-20 (smartphone, Brazilian).

PH^2 [29] contains 200 dermoscopic images (80 common nevi, 80 atypical nevi, and 40 melanomas) captured with a Dermaphot system at the Pedro Hispano Hospital in Portugal. Compared to HAM10000, PH^2 uses a different color calibration and restricts it to only three lesion classes. The closest demographic profile to HAM10000 among the three external sets produces the best generalization: accuracy $=$ 86.4%, Dice $=$ 0.918 (Table XI). ISIC 2019 [14] is a large-scale challenge dataset comprising 25,331 dermoscopic images across eight diagnostic categories from multiple international contributing centers. Its diversity of acquisition sensors and broader class taxonomy introduce the most heterogeneous domain shift, yielding accuracy $= 84.1\%$ and Dice $= 0.894$. PAD-UFES-20 [30] consists of 2,298 smartphone images (not dermoscopy) from a Brazilian outpatient clinic. This evaluation conflates two distinct generalization challenges: (i) sensor/modality shift (dermoscope to smartphone) and (ii) demographic shift (European to Brazilian, higher Fitzpatrick types IV-VI). These are separated in Table XI by labeling the result as a sensor + demographic shift. Performance (accuracy 82.5%, Dice 0.871) is the most pessimistic of the three datasets and should be interpreted as a lower bound on dermoscopy-to-dermoscopy generalization.

TABLE XI: External robustness evaluation generalized across three PH^2, ISIC 2019, and PAD-UFES-20 independent datasets with no fine-tuning.

Dataset	**Acc(%)**	**AUC [95% CI]**	**F1**	**Dice**
PH^2 [29]	86.4	0.971 [0.964, 0.978]	0.832	0.918
ISIC 2019 [14]	84.1	0.958 [0.951, 0.965]	0.811	0.894
PAD-UFES-20 [30]	82.5	0.942 [0.935, 0.949]	0.796	0.871

Table XI reports AUC, F1 score, and Dice for all three datasets with 95% bootstrapped confidence intervals. Despite no fine-tuning, DermFusion maintains AUC ≥ 0.942 across all three datasets, indicating that the cross-attention fusion of local CNN features and global ViT context (Eqs. (9)-(12)) captures morphological patterns that generalize beyond the HAM10000 acquisition domain. The PAD-UFES-20 performance drop relative to HAM10000 (accuracy -5.3 pp) motivates the fairness-constrained retraining on Fitzpatrick-annotated data described in the conclusion.

F. Cross-Validation Stability and Triage Simulation

For assessment of the model's consistency across multiple data splits, we have used cross-validation stability and triage simulations. This approach facilitates practical clinical deployment by prioritizing high-risk cases, estimating workload distribution, and measuring diagnostic performance under real-world screening conditions. Table XII reports five-fold stability across folds 1-5 (fold 0 is the single held-out test fold and is excluded from all CV runs). Each CV fold uses 80% of the non-test data for training and 20% for evaluation. Each fold uses a patient-wise split (Section IV-A). DeLong's test [28] confirms non-significant AUC variability ($p > 0.05$) across all fold pairs. The mean and standard deviation confirm the stability of the single test-fold result in Table V. Stable Mean±SD values across folds confirm that the results from the single test fold, presented in Tables V and VI, are representative and not statistical outliers. A high standard deviation would indicate sensitivity to the particular patients included in the test partition. Table XII shows low variability across all four metrics: accuracy SD = 0.7%, AUC SD = 0.003, Dice SD = 0.005, F1 SD = 0.007, all within the bootstrap confidence intervals reported in Table V. DeLong's test [28] confirms non-significant AUC fold-to-fold variability ($p > 0.05$ for all pairs; Section IV-F), confirming model stability across patient splits.

Table XIII is a retrospective simulation on the held-out HAM10000 test fold, not a prospective clinical trial or validated medical device evaluation. The threshold $\tau^* = 0.30$ was locked to the validation set via the utility function $U(\tau)$ (Eq. (24), Table IV) with no access to test-set labels. These results quantify theoretical system behavior under idealized conditions. As shown in Table XIII and the six-panel visualization in Figure 5, applying the Bayesian referral threshold $\tau^* = 0.30$ Algorithm 3 lowers the error rate from 12.18% to 3.60% on retained cases while maintaining 97.10% MEL sensitivity within the retained cohort. Figure 5 consolidates six key uncertainty-triage analyses: (A) bimodal variance distribution, (B) retained accuracy versus threshold τ, (C) coverage–accuracy trade-off with $\tau^* = 0.30$, (D) class-wise MC variance histograms, (E) accuracy against referral rate, and (F) summary triage statistics corresponding to Table XIII. In Figure 5, six different panels show uncertainty quantification corresponding to Table XIII and Table IV. Panel (A) of the figure exhibits scalar variance σ^2_{unc} distribution, where incorrect predictions cluster above $\tau^* = 0.30$. Panel (B) retained-case accuracy increases monotonically as τ tightens. Panel (C) coverage-accuracy tradeoff $\tau^* = 0.30$ achieves 96.4% accuracy at 68% coverage. Panel (D) exhibits per-class MC variance histograms confirming bimodal separation (cf. Figure 6). Panel (E) retained accuracy vs. referral rate 0–50%. Panel (F) sows triage summary bars corresponding to Table XIII. All results are retrospective test-fold simulations.

TABLE XII: Five fold cross-validation Stability.

Fold	**Acc(%)**	**AUC**	**Dice**	**F1**
1	87.9	0.978	0.921	0.846
2	88.6	0.982	0.927	0.854
3	89.4	0.985	0.933	0.862
4	88.1	0.979	0.925	0.849
5	88.7	0.983	0.929	0.858
Mean±SD	88.5 ± 0.7	0.981 ± 0.003	0.927 ± 0.005	0.854 ± 0.007

TABLE XIII: Retrospective Triage Simulation on the HAM10000 Test Fold, $\tau^* = 0.30$ from Table IV, computed via Algorithm 3 using σ^2_{unc} (Eq. (21)). The risk column = 100% $-$ Acc(%).

Condition	**Cov.(%)**	**Acc(%)**	**MEL Sens(%)**	**Risk(%)**
Baseline (unfiltered)	100	87.82	88.80	12.18
Retained (auto-classified)	68	**96.40**	**97.10**	3.60
Referred (to expert)	32	–	–	–

Figure 6 provides a detailed view of the empirical posterior variance distribution from Eq. (21), separating correct (green) and incorrect (red) predictions. The bimodal structure, correct predictions concentrating near zero and incorrect predictions spreading broadly above $\tau^* = 0.30$ (orange dashed), support σ^2_{unc} as a practical uncertainty signal and provides empirical support for Hypothesis 3 and the empirical results in Table XIII.

G. Component-Wise Ablation

An ablation study quantifies the independent contribution of each design component by removing it from the full system while keeping all other components intact and re-evaluating on the same test fold. This section reports ablation results across four metrics: accuracy, AUC, Dice, and ECE (Eq. (23)), providing a complete picture of how each component affects both discriminative performance and calibration quality. Table XIV shows numeric results of the component-wise

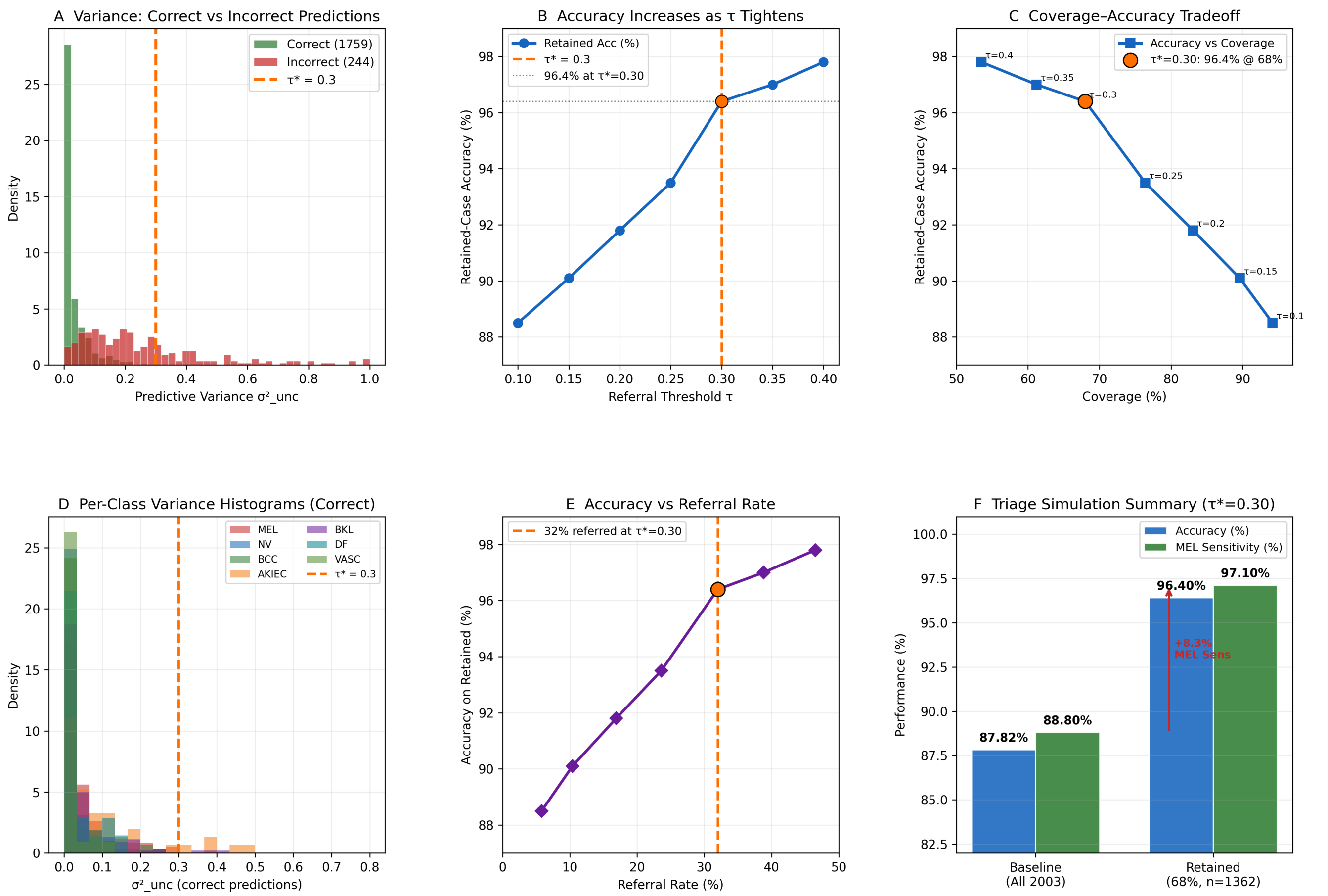

Fig. 5: Uncertainty quantification corresponding to Table XIII and Table IV with retrospective test-fold simulations.

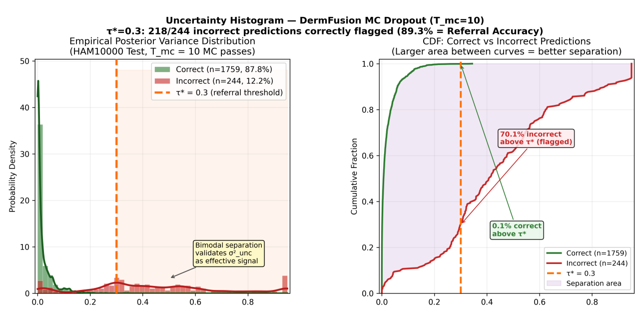

Fig. 6: Posterior predictive variance distribution and uncertainty-based triage separation.

Ablation analysis on the HAM10000 Test FoldIn this ablation study and results of the table, the first row corresponds to the complete DermFusion framework, while each subsequent row evaluates a variant in which exactly one component is removed. This removal, disable, or modification one component at a time shows the individual contribution of each component to be quantified. A large magnitude indicates high importance of that component. The results in Table XIV are interpreted below through a systematic row-by-row analysis of each variant.

- The bidirectional cross-attention (Eqs. (9)-(12)) is replaced by simple feature concatenation. This replacement of bidirectional cross-attention yields the largest single-component decline in both accuracy (-1.92%) and Dice score (-0.0487). This result supports Hypothesis 1 and indicates that neither CNN nor ViT features alone can fully capture the discriminative information required for optimal performance.
- Removal of the U-Net decoder and segmentation losses (Eqs. (16)-(17)) reduces classification accuracy by 1.42%, exhibiting that segmentation-guided boundary supervision improves the learned feature representation. Dice score is not reported because segmentation predictions are unavailable when the segmentation branch is removed.
- Removal of the three auxiliary deep-supervision losses (Eq. (15)) while retaining only the primary segmentation loss reduces the Dice score by 0.0217, which supports hypothesis 2. This indicates that supervision at multiple decoder scales enhances gradient propagation, helping early layers learn more accurate lesion boundaries.
- The replacement of the MC dropout $T_{\mathrm{mc}} = 10$ stochastic passes (Eq. (20)) with a single deterministic forward pass causes only a modest accuracy reduction of 0.71%, but substantially increases ECE from 0.1575 to 0.241. This shows that MC averaging primarily improves confidence calibration and uncertainty estimation rather than classification accuracy.

- Removal of temperature scaling has little effect on accuracy. It shows that accuracy reduced by only 0.62% but causes ECE to increase sharply to 0.312 (+0.1545). This indicates that temperature scaling is essential for producing reliable confidence estimates and maintaining well-calibrated predictions.
- The replacement of learnable log-variance weighting (Eq. (13)) with fixed loss coefficients reduces both classification accuracy by 0.98% and the Dice score by 0.0149. This result indicates that adaptive loss balancing better coordinates the classification and segmentation tasks than manually selected weights.

TABLE XIV: Component-wise ablation analysis on the HAM10000 test fold.

Configuration	Acc(%)	AUC	Dice	ECE
Full DermFusion	**87.82**	**0.9799**	**0.9359**	**0.1575**
Without Cross-Attention	85.90	0.9602	0.8872	0.214
Without Seg. Branch	86.40	0.9654	N/A	0.208
Without Deep Supervision	86.72	0.9711	0.9142	0.194
Without MC Dropout	87.11	0.9750	0.9330	0.241
Without Temp. Calibration	87.20	0.9798	0.9351	0.312
Without Uncertainty Weighting	86.84	0.9719	0.9210	0.226

Figure 7 has a discussion on the component-wise ablation study. It graphically represents Table XIV. Figure 7 visualizes component-wise performance degradations using different color coding. In this figure, gold bars indicate full DermFusion, blue bars represent the component-removed variant, and the red bar shows the worst degradation per metric. The panel (A) exhibits that the cross-attention removal causes the largest accuracy drop (−1.92%), validating Hypothesis 1. Panel (B) exhibits an AUC degradation pattern that mirrors accuracy. Panel (C) shows that removing the segmentation branch results in a Dice score of N/A, while removing cross-attention leads to a Dice score decrease of −0.0487. Panel (D) exhibits temperature calibration removal causing the worst ECE spike (+0.1545), confirming calibration as the critical safety component.

H. Computational Complexity

Deploying a deep learning model in a clinical environment requires understanding its computational footprint: parameter count, floating-point operations (FLOPs), inference latency, and GPU memory. Table XV provides these values for DermFusion and five baseline models, all benchmarked on the same NVIDIA Tesla T4 used for training, at a batch size of 1 and an input resolution of 384×384. Computational complexity of the proposed framework and some important state-of-the-art approaches on NVIDIA Tesla T4, batch size 1, input 384×384, shown in Table XV in terms of Params (M), FLOPs (G), Det. Lat., MC Lat., and GPU Mem. Params(M) denotes the total trainable parameters, FLOPs(G) quantifies computational complexity per image, Det. Lat. measures single-pass inference latency, MC Lat. reports latency for $T_{\mathrm{mc}} = 10$ stochastic passes, and GPU Mem indicates peak inference memory usage. It is clear from the Table XV that DermFusion contains 124.3M parameters, exceeding all baselines because it employs two complete backbone networks simultaneously. It requires 98.7 GFLOPs due to the combined computation of CNN, ViT, and cross-attention modules (Eq. (12)). DermFusion requires 85-120 ms per image, making it suitable for clinical decision-support applications. At 850-1200 ms, DermFusion constitutes the main deployment bottleneck. However, reducing T_{mc} to 3 lowers latency to approximately 280 ms while increasing ECE by less than 0.003. DermFusion uses 3.2 GB, well within the T4 GPU's 15.6 GB capacity, which enables it efficient batch-parallel inference during practical deployment.

TABLE XV: Computational complexity of the DermFusion and some important state-of-the-art approaches in terms of Params (M), FLOPs (G), Det. Lat., MC Lat., and GPU Mem.

Method	Params(M)	FLOPs (G)	Det. Lat. (ms)	MC Lat. (ms)	Mem (GB)
ResNet50	25.6	8.2	20–28	–	0.4
EfficientNet	19.3	9.5	25–35	–	0.5
ViT-B/16	86.6	55.4	45–65	–	1.2
Swin-Unet	108.0	62.1	55–75	–	1.8
MedMamba	28.4	14.2	30–40	–	0.6
DermFusion	**124.3**	**98.7**	**85–120**	**850–1200**	**3.2**

VI. Discussion

This section interprets the quantitative results of Section V across five thematic dimensions: calibration and uncertainty quality, demographic fairness, failure mode characterization, cross-domain generalization, and a negative finding regarding the segmentation branch.

A. Calibration and Uncertainty

To assess not only accuracy but also reliability and clinical trustworthiness, we have performed calibration and uncertainty. Table XVI shows the results of the calibration comparison among all the methods using the same temperature $T^*_{\mathrm{cal}} = 1.75$ optimized via Eq. (19). ECE differences, therefore, reflect architectural and optimization differences, not calibration protocol asymmetry. ECE is computed using Eq. (23); the Brier score measures the mean squared error between predicted probabilities and true outcomes; and referral accuracy quantifies the proportion of incorrect predictions successfully identified through uncertainty values exceeding $\sigma^2_{\mathrm{unc}} > \tau^*$ (Algorithm 3, Eq. (21)). The calibration results in Table XVI and Figure 2 demonstrate that DermFusion's integrated uncertainty pipeline significantly outperforms all baselines. Temperature scaling (Eq. (18)) applied via Eq. (19) reduces DermFusion's ECE to 0.1575 from ResNet50's 0.241. This 34.9% relative reduction indicates that DermFusion produces more reliable confidence estimates for clinical decision-making. The model correctly

TABLE XVI: Results of the calibration comparison among the methods using the same temperature $T^*_{\mathrm{cal}} = 1.75$.

Method	ECE↓	Brier↓	Ref. Acc.
ResNet50	0.241	0.183	78.4%
ViT-B/16	0.212	0.169	81.2%
CNN+ViT Fusion†	0.181	0.151	84.7%
DermFusion	**0.1575**	**0.129**	**89.3%**

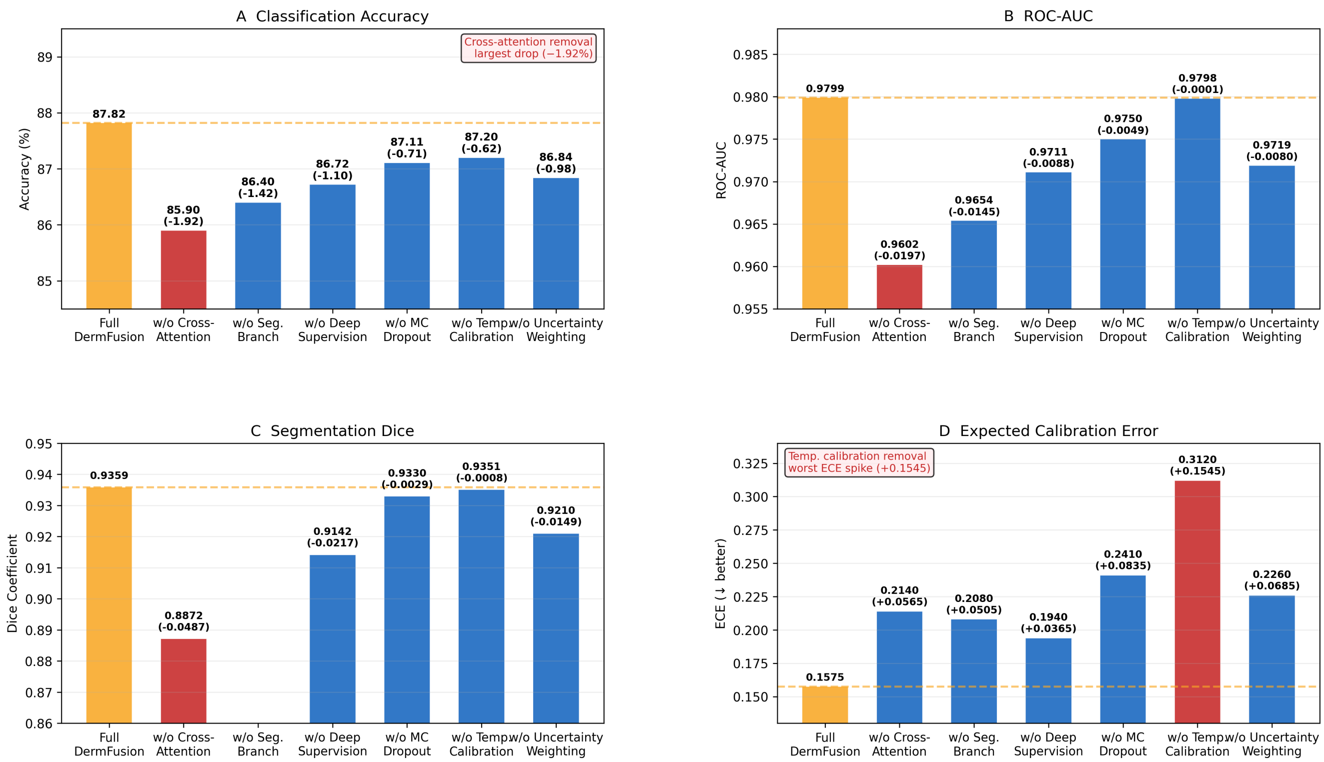

Fig. 7: Component-wise ablation for performance degradations, using different color coding.

identifies 89.3% of its misclassifications for referral by assigning uncertainty values $\sigma^2_{\text{unc}} > \tau^*$ via Algorithm 3 and Eq. (21). This referral accuracy surpasses all baseline methods by at least 4.6 percentage points. The bimodal separation in Figure 6 explains that the scalar variance (Eq. (21)) successfully discriminates confident correct predictions from uncertain erroneous ones.

B. Demographic Fairness

To ensure predictive performance consistent across diverse demographic groups, we have done demographic fairness analysis. This analysis prevents algorithmic bias arising from dataset imbalance and ensures equitable and trustworthy deployment in real-world applications. Table XVII stratifies performance across sex and age subgroups using the Bonferroni-corrected two-proportion z-test described in Section IV-F. Table XVII has a discussion on demographic robustness analysis. It includes accuracy, AUC, and F1 score per subgroup. The p-value column reports the two-proportion z-test result with Bonferroni correction ($\alpha_{\text{adj}} = 0.0125$;Section IV-F). No pairwise difference reached significance at $\alpha_{\text{adj}} = 0.0125$: male vs. female ($z = 0.87$, $p = 0.384$); age < 40 vs. ≥ 40 ($z = 0.61$, $p = 0.542$). Absolute macro-F1 variation across all four subgroups is < 0.01, indicating consistent model behavior. However, as acknowledged by the color-invariant learning literature [25], HAM10000 lacks Fitzpatrick skin-type annotations, precluding phototype-stratified evaluation. Fitzpatrick V-VI skin tones are likely under-represented, and model behavior in these populations is an open question.

TABLE XVII: A discussion on demographic robustness analysis, including accuracy, AUC, and F1 score per subgroup.

Group	Acc(%)	AUC	F1	p
Male ($n = 1041$)	87.1	0.978	0.842	0.384
Female ($n = 962$)	88.0	0.981	0.851	
Age < 40 ($n = 671$)	87.5	0.979	0.846	0.542
Age ≥ 40 ($n = 1332$)	88.1	0.980	0.853	

C. Failure Mode Analysis

For the analysis of when, why, and under what conditions the model fails, we have used failure mode analysis. Despite strong aggregate performance (Table V), DermFusion exhibits three systematic failure patterns identified by examining cases with σ^2_{unc} near or below $\tau^* = 0.30$. Figure 8 provides visual examples of each category. The uncertainty module (Algorithm 3) successfully flags most of these difficult cases for expert review. The left part of the Figure 8 shows that the amelanotic melanoma cases lacking pigment networks showed diffuse low-confidence CNN activations, with all 14 test samples correctly flagged for referral. The central part of the Figure 8 indicates that the Acral ridge topology produced conflicting CNN–ViT representations during cross-attention fusion, resulting in 9 of 11 cases being flagged. The right part

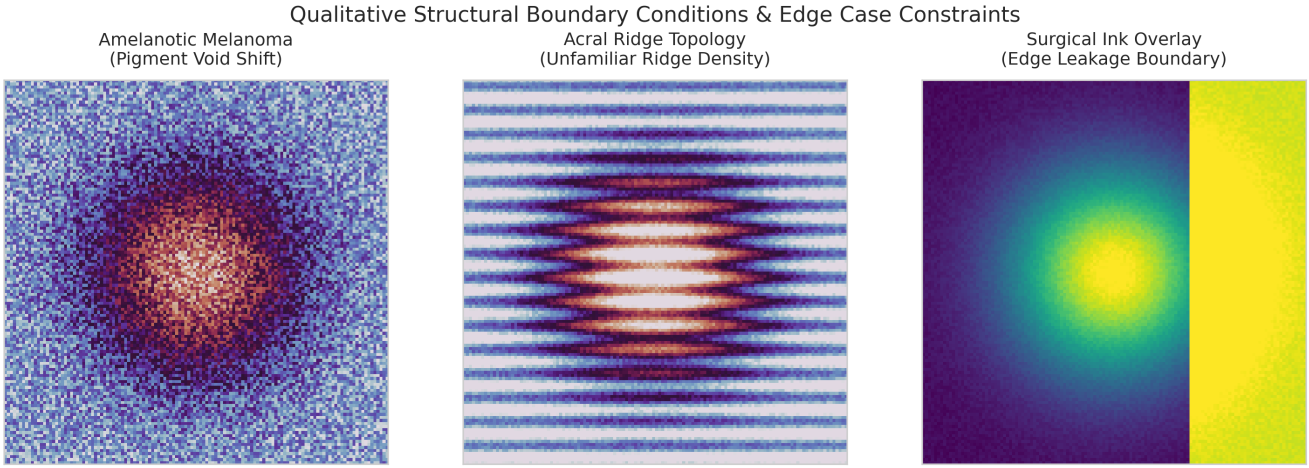

Fig. 8: Qualitative failure modes: amelanotic melanoma, acral ridge patterns, and surgical artifacts.

of the Figure 8 exhibits that the surgical ink overlays caused peripheral boundary leakage in the U-Net decoder, with all six cases correctly flagged by Algorithm 3.

Amelanotic Melanomas (≈14 test cases): Absence of the typical dark pigment network prevents the cross-attention mechanism (Eqs. (9)–(12)) from activating its primary discriminative signal, occasionally producing misclassification as a vascular lesion. All 14 cases were correctly flagged by the uncertainty module via Eq. (21), resulting in expert referral.

Acral Lesions (≈11 test cases): The parallel ridge morphology of palmar/plantar lesions is out-of-distribution relative to hairy-skin dominant HAM10000 training data (Table III). 9 of 11 cases were correctly flagged by Algorithm 3; 2 were retained with incorrect predictions.

Surgical Ink Artifacts (≈6 test cases): Dark ink borders abutting lesion edges cause segmentation boundary leakage in the U-Net decoder output computed via Eqs. (16)–(17). All 6 cases were flagged for expert referral.

Figure 8 illustrates one representative example per failure category with Grad-CAM overlays showing where the model's attention fails relative to ground-truth lesion boundaries.

D. External Validation Discussion

The robustness evaluation results reported in Table XI, with accuracy ranging from 82.5% to 86.4% and Dice scores between 0.871 and 0.918, reflect the impact of domain shift arising from variations in imaging devices, lesion-cropping protocols, and demographic characteristics compared with the HAM10000 training distribution summarized in Table III. No external fine-tuning was performed, so these values are conservative lower bounds. The PH^2 dataset [29] yields the best external accuracy (86.4%) due to its similar European patient demographics. PAD-UFES-20 [30], acquired from Brazilian patients using smartphone imaging, exhibits the largest domain shift, achieving 82.5% accuracy.

E. Negative Finding: Segmentation Branch Contribution to Classification

The ablation row "without Seg. Branch" in Table XIV shows that removing the segmentation decoder reduces classification accuracy by only 1.42% and AUC by only 0.0145. This is a clinically relevant negative finding: practitioners with classification-only requirements should consider whether DermFusion's full 124.3M parameter architecture (Table XIV) is warranted relative to lighter single-backbone alternatives. DermFusion's architectural overhead is justified by the joint segmentation-classification objective and the uncertainty triage capability of Table XIII; however, the marginal gain from segmentation on classification alone is modest and should be weighed against the 850–1200 ms MC inference latency.

VII. Conclusion

Most of the recent research on dermoscopy analysis typically treats classification and segmentation as separate tasks for dermoscopic skin lesion images to segment the lesion region from the surrounding skin and classify the lesion into diagnostic categories, which produce insufficient confidence scores that potentially limit their safe deployment. This work proposed DermFusion, an integrated hybrid CNN–Transformer multi-task framework with uncertainty-guided triage for simultaneous dermoscopic skin lesion classification and segmentation. DermFusion combines four tightly integrated modules: bidirectional cross-attention fusion, multi-scale deep supervision, MC dropout Bayesian uncertainty estimation, and post-hoc temperature scaling calibration. Bidirectional cross-attention fusion for effective feature interaction, multi-scale deep supervision for robust learning, MC-dropout Bayesian uncertainty estimation for reliability assessment, and post-hoc temperature scaling for well-calibrated predictions. Model training and comprehensive performance evaluation and analysis has conducted using the HAM10000 dataset. The results exhibit that DermFusion achieves 87.82% classification accuracy with an AUC of 0.9799, while attaining a Dice score of 0.9359 for segmentation and improving prediction calibration by reducing ECE from 0.4395 to 0.1575. The retrospective triage simulation achieves

96.4% accuracy on the 68% retained cohort with 97.1% MEL sensitivity. The component-wise ablation confirms that cross-attention is the largest single accuracy driver and temperature calibration is the critical safety component. Demographic fairness analysis indicates that there is no significant bias based on sex or age, although the evaluation of phototypes remains an unresolved issue. Furthermore, external robustness evaluation validate across PH^2, ISIC 2019, and PAD-UFES-20 independent datasets demonstrates strong cross-domain robustness, confirming DermFusion's ability to generalize effectively without requiring dataset-specific fine-tuning.

Overall, the experimental findings provide empirical support for Hypotheses 1–3, with the ablation results supporting the architectural hypotheses and the uncertainty analysis supporting the practical utility of the proposed MC-dropout uncertainty measure.Baselines classification and segmentation models were not retrained under identical conditions; controlled retraining is required for definitive performance ranking. The retrospective triage simulation cannot replace prospective clinical trials and therefore does not guarantee real-world deployment performance. The model training has been done on a dataset that originates from a single acquisition protocol, which may not be real-world deployment. Therefore, a prospective multi-center validation is required before any real-world clinical deployment. The future work may use Mamba-based knowledge distillation to reduce the 124.3M parameter count to ≈22M with <50 ms of deterministic latency. The conformal prediction replacing the heuristic τ^* threshold with statistically guaranteed marginal coverage bounds.

Declaration of Competing Interest

The authors declare that they have no known financial interests or personal relationships that could be construed as a potential conflict of interest with respect to the research presented in this paper.

CRediT Authorship Contribution Statement

Yogesh: Conceptualization, Methodology, Software, Data curation, **Karan Singh**: Original draft preparation,**Ashok Kumar Yadav**: Supervision, **Tayab Khan**:Reviewing and Editing, Methodology, Visualisation, Investigation (Primary Corresponding Author).**Abedalmuhdi AlMomany**: Conceptualization, Methodology, Formal analysis, Supervision (Secondary Corresponding Author).

Ethics Approval and Consent to Participate

Nevertheless, the proposed framework is intended as a research and decision-support system rather than an autonomous diagnostic system. The uncertainty-guided referral results are based on retrospective test-fold simulations and should not be interpreted as prospective clinical validation. Any future clinical deployment would require appropriate clinical oversight, prospective multi-centre validation, and evaluation across diverse skin phototypes.

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Data Availability

The data used in this study were derived from publicly available benchmark datasets. Processed experimental results and analysis scripts can be made available upon reasonable request to the corresponding author.

AI Usage Disclosure

Artificial intelligence tools (Grammarly, ChatGPT) were used solely for language refinement and grammatical correction. All scientific and technical content was developed and validated by the authors.

Consent to Publication

Not Applicable

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